

DAY 1 • LESSON 3-1

Understand Ratios

Maryland Standard 6.AT.1 • Show your work, then write your answer on the answer line.

REMEMBER

review this before you start

- A ratio compares two amounts. Order matters! Write numbers in the exact same order as the words.
- Part-to-Part: Compares one piece to another piece (e.g., 3 red to 2 blue = 3 : 2).
- Part-to-Whole: Compares one piece to the total amount (e.g., 3 red to 5 total = 3 : 5). Find the total first!

EXERCISES

1. A fruit smoothie recipe uses 3 cups of strawberries, 2 cups of bananas, and 5 cups of milk (10 cups total smoothie).

a. Ratio of strawberries to bananas (using the word 'to'): _____

b. Ratio of strawberries to bananas (using a colon ':'): _____

c. Ratio of bananas to the **total** cups of smoothie: _____

d. Is question (c) part-to-part or part-to-whole?

☐ Part-to-part

☐ Part-to-whole

2. Tahira makes salad dressing using 1 spoon of vinegar for every 3 spoons of olive oil. Fill in the missing numbers in the ratio table:

Vinegar (spoons)	1	2	3	4	5
Olive Oil (spoons)	3	6			

3. A snack mix uses 2 cups of pretzels for every 5 cups of cereal. Marcus uses 6 cups of pretzels. How many cups of cereal does he need?

• Step 1: Pretzels increased from 2 to 6. Multiply factor: $2 \times \underline{\hspace{1cm}} = 6$

• Step 2: Multiply cereal by the same number: $5 \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

Final Answer _____ cups of cereal

CHECK YOURSELF

☐ Solved on my own

☐ Needed a hint

☐ Need help on Question # _____

DAY 2 • LESSON 3-2

Rates and Unit Rates

Maryland Standard 6.AT.2 • Show your work, then write your answer on the answer line.

REMEMBER

review this before you start

- A rate compares two quantities with different units (like miles and hours, or dollars and days).
- A unit rate tells the amount for 1 unit. Divide the requested amount by the number of units (e.g., $\$15 \div 3 = \5 each)!

EXERCISES

1. A commuter metro train travels 80 miles in 2 hours. What is the train's unit speed in miles per hour?

- Divide miles by hours: $80 \text{ miles} \div 2 \text{ hours} = \underline{\hspace{2cm}}$ miles in 1 hour

Unit Speed _____ miles per hour (mph)

2. During gym class, Alex runs 12 laps in 3 minutes. What is Alex's running rate in laps per minute?

- Divide laps by minutes: $12 \text{ laps} \div 3 \text{ minutes} = \underline{\hspace{2cm}}$ laps in 1 minute

Unit Rate _____ laps per minute

3. Yelina compares two ticket options for riding the metro train:

- Option A (Day Pass): Costs \$6.00 for 1 day.
- Option B (5-Day Weekday Pass): Costs \$25.00 for 5 days.

a. Find the cost per day for Option B ($\$25.00 \div 5$): \$ _____ per day

b. Which pass is cheaper per day?

☐ Option A	☐ Option B
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4. Yelina needs to ride the train on 4 days this week. Which option should she buy?

- 4 single day passes cost: $4 \times \$6.00 = \$ \underline{\hspace{2cm}}$
- The 5-day pass costs: \$25.00

Which choice saves Yelina money? _____

CHECK YOURSELF

☐ Solved on my own	☐ Needed a hint	☐ Need help on Question # _____
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DAY 3 • LESSON 3-3

Equivalent Ratios Using Tables

Maryland Standard 6.AT.3a · Show your work, then write your answer on the answer line.

REMEMBER

review this before you start

- A ratio table organizes equivalent pairs. Multiply or divide both numbers by the same number to find missing values.
- Rule: What you do to the top row, you must do to the bottom row!

EXERCISES

1. To make craft clay, an art teacher mixes 2 cups of flour for every 1 cup of salt. Fill in the missing values in the table:

Flour (cups)	2	4	6	8	10
Salt (cups)	1	2			

2. An artist mixes custom orange paint with 4 cans of red paint for every 3 cans of yellow paint. Fill in the missing numbers:

Red Paint (cans)	4	8		16	20
Yellow Paint (cans)	3	6	9		15

3. The artist wants to use 28 cans of red paint. How many cans of yellow paint are needed?

- Think: What multiplies 4 to make 28? $4 \times \underline{\hspace{1cm}} = 28$
- Multiply yellow paint (3) by that same number: $3 \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

Yellow Cans Needed _____ cans

4. Deon compares two bottles of dish soap: Bottle A is 20 oz for \$4, and Bottle B is 40 oz for \$7.

- Double Bottle A to 40 oz: $\$4 \times 2 = \$\underline{\hspace{1cm}}$ for 40 oz.
- Bottle B costs: \$7.00 for 40 oz.

Which bottle is the better deal for 40 oz? Bottle _____

CHECK YOURSELF

☐ Solved on my own

☐ Needed a hint

☐ Need help on Question # _____

DAY 4 • LESSON 3-4

Equivalent Ratios Using Graphs

Maryland Standard 6.AT.3a · Show your work, then write your answer on the answer line.

REMEMBER

review this before you start

- Ordered Pairs (x, y): In a point (x, y), x goes across (horizontal) and y goes up (vertical).
- Straight Line through (0, 0): All equivalent ratios on a graph form a straight line passing through the origin (0, 0)!

EXERCISES

1. A coach packs soccer balls into bags. Each bag holds 5 soccer balls. Let x = bags and y = balls. Complete the table and write each ordered pair (x, y):

Bags (x)	Soccer Balls (y)	Ordered Pair (x, y)
1	5	(1 , 5)
2	10	(2 , 10)
3		(3 , _____)
4		(4 , _____)

2. What does the coordinate point (0, 0) mean in this soccer bag problem?
Meaning of (0, 0): 0 bags hold _____ soccer balls.
3. Lucy strings charm bracelets. For every 1 charm, she uses 4 beads.
 - a. If Lucy uses 3 charms, how many beads does she need? ($3 \times 4 =$ _____)
Write the ordered pair (charms, beads): (3 , _____)
 - b. If the graph shows the point (6, 24), what does this mean in words?
Meaning: 6 charms and _____ beads.
4. A classmate claims that the point (5, 22) belongs on Lucy's bracelet line.
 - Check: 5 charms $\times$ 4 beads per charm = _____ beads.
 Is (5, 22) on the line?

☐ Yes	☐ No
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Circle one.

What is the correct point for 5 charms? (5 , _____)

CHECK YOURSELF

☐ Solved on my own	☐ Needed a hint	☐ Need help on Question # _____
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DAY 5 • LESSON 3-5

Comparing Ratio Relationships

Maryland Standard 6.AT.3 • Show your work, then write your answer on the answer line.

REMEMBER

review this before you start

- To compare two ratios fairly, make one quantity the exact same number in both ratios!
- Once one number matches, compare the other number to see which is greater, bluer, or gives more points.

EXERCISES

1. Two painters mix purple paint with different recipes:

- Reginald: 3 red cans for every 4 blue cans (3 : 4)
- Anwar: 2 red cans for every 3 blue cans (2 : 3)

Make the red paint equal (6 red cans for both):

- Reginald (multiply by 2): 6 red cans and _____ blue cans.
- Anwar (multiply by 3): 6 red cans and _____ blue cans.

Whose paint mixture will look bluer? _____'s mixture

2. Brielle and Andrew design video game reward points:

- Brielle's Game: awards 2 stars for every 5 levels completed.
- Andrew's Game: awards 3 stars for every 6 levels completed.

Scale both games to 30 levels completed:

- Brielle (multiply by 6): $5 \times 6 = 30$ levels $2 \times 6 =$ _____ stars.
- Andrew (multiply by 5): $6 \times 5 = 30$ levels $3 \times 5 =$ _____ stars.

Which game awards more stars for 30 levels? Game _____

3. Two friends compare their babysitting earnings:

- Claire earned \$30 for 3 hours: $\$30 \div 3 = \$$ _____ per hour.
- Nakai earned \$44 for 4 hours: $\$44 \div 4 = \$$ _____ per hour.

Who earned a higher hourly rate? _____

CHECK YOURSELF

☐ Solved on my own	☐ Needed a hint	☐ Need help on Question # _____
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DAY 6 • LESSON 3-6

Converting Within One System

Maryland Standard 6.AT.3 • Show your work, then write your answer on the answer line.

REMEMBER

review this before you start

- Converting means writing the same amount in a different unit.
- Customary: 1 foot = 12 inches | 1 pound = 16 ounces | 1 quart = 32 fl oz
- Metric: 1 meter = 100 centimeters | 1 liter = 1,000 milliliters (mL)

EXERCISES

- 1.** A roller coaster requires riders to be at least 48 inches tall. Caleb is 4 feet tall.

- Convert Caleb's height to inches (1 foot = 12 inches):

4 feet $\times$ 12 inches per foot = _____ inches tall.

Is Caleb tall enough to ride?

☐ Yes

☐ No

Circle one.

Explain: Because Caleb is _____ inches tall, which meets the 48-inch requirement.

- 2.** Diamond wants to drink 2 liters of water today. Her water bottle holds 500 milliliters (mL).

- a.** Convert 2 liters to milliliters (1 liter = 1,000 mL):

2 liters $\times$ 1,000 mL/L = _____ mL

- b.** How many full 500 mL bottles must she drink to reach 2,000 mL?

2,000 mL $\div$ 500 mL per bottle = _____ full bottles

- 3.** When Marcus converts 3 feet into 36 inches, the number got bigger (3 to 36), but Marcus stayed the same height! Why did the number get bigger?

Circle the best explanation:

- a.** Inches are smaller than feet, so it takes more inches to measure the same length.
- b.** Marcus grew taller during the math lesson.

Correct Choice (A or B) _____

CHECK YOURSELF

☐ Solved on my own

☐ Needed a hint

☐ Need help on Question # _____

DAY 7 • LESSON 3-7

Converting Between Systems

Maryland Standard 6.AT.3 • Show your work, then write your answer on the answer line.

REMEMBER

review this before you start

- Use $\approx$ with the rounded conversion factors below; some conversions, such as 1 in = 2.54 cm, are exact.
- Distance: 1 kilometer $\approx$ 0.6 mile
- Weight: 1 pound $\approx$ 0.45 kilogram

EXERCISES

- 1.** In Canada, a highway speed limit sign says 100 km/h (kilometers per hour).

a. Convert 100 km/h to miles per hour using 1 km $\approx$ 0.6 miles:

$100 \times 0.6 \approx$ _____ miles per hour (mph)

b. Near her home in the US, the speed limit is 65 mph. Is 100 km/h faster or slower than 65 mph?

☐ Faster

☐ Slower

Circle one.

- 2.** A puppy at the vet weighs 20 pounds. The medicine chart lists doses in kilograms (kg).

a. Convert 20 pounds to kilograms using 1 pound $\approx$ 0.45 kilogram:

$20 \times 0.45 \approx$ _____ kilograms (kg)

b. The medicine dose is 1 chewable tablet for every 3 kg of body weight.

In this math model, how many tablets does the chart indicate for 9 kg? $(9 \div 3) =$ _____ tablets

- 3.** Why do mathematicians write 1 pound $\approx$ 0.45 kilogram with the wavy sign ($\approx$) instead of an equals sign (=)?

Circle the best explanation:

a. 0.45 is a rounded, approximate number, so 1 lb and 0.45 kg are very close but not exact.

b. The veterinarian made a calculation mistake.

Correct Choice (A or B) _____

CHECK YOURSELF

☐ Solved on my own

☐ Needed a hint

☐ Need help on Question # _____